

MATH3881/5881: Statistical Machine Learning Theory

Week 3: Supplement: Convergence Proofs and Convexity of ML Models

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Optional reading. This supplement is optional material for students who want the full derivations underlying Section 2 of the Week 3 lecture notes. It contains proofs of the Descent Lemma and the two gradient descent convergence theorems, and Hessian-based proofs that logistic and softmax regression are convex. None of this material is examinable beyond what is stated as theorems in the main notes; students may safely skip it.

The Week 3 lecture notes state three central results without proof, all in Section 2 (Convex Optimisation in Machine Learning):

- the *Descent Lemma* (the quadratic upper bound for L -smooth objectives);
- the $O(1/t)$ convergence rate of gradient descent on convex L -smooth objectives;
- the linear rate $(1 - \mu/L)^t$ for μ -strongly convex L -smooth objectives.

Sections 1–3 of this supplement give the proofs. Sections 4 and 5 then show that two further models from Week 1, logistic and softmax regression, fit the convex framework: their training losses are convex in the parameters, and we identify a smoothness constant L for each. The main notes prove this only for linear regression.

Throughout, we follow the notation of the Week 3 notes: $C : \mathbb{R}^d \rightarrow \mathbb{R}$ is the training loss, θ is the parameter vector, ∇C and $\nabla^2 C$ are the gradient and Hessian, and $\succeq 0$ denotes positive semidefiniteness.

1. Proof of the Descent Lemma

We restate the Descent Lemma from the Week 3 notes for convenience.

Descent Lemma. *Let $C : \mathbb{R}^d \rightarrow \mathbb{R}$ be L -smooth. Then for any $\theta, \phi \in \mathbb{R}^d$,*

$$C(\phi) \leq C(\theta) + \nabla C(\theta)^\top (\phi - \theta) + \frac{L}{2} \|\phi - \theta\|^2.$$

Proof: *Step 1.* (Fundamental theorem of calculus along the segment from θ to ϕ .)

$$C(\phi) - C(\theta) = \int_0^1 \nabla C(\theta + t(\phi - \theta))^\top (\phi - \theta) dt.$$

Step 2. (Subtract the linear term.) Because $\nabla C(\theta)^\top(\phi - \theta)$ does not depend on t ,

$$C(\phi) - C(\theta) - \nabla C(\theta)^\top(\phi - \theta) = \int_0^1 [\nabla C(\theta + t(\phi - \theta)) - \nabla C(\theta)]^\top(\phi - \theta) dt.$$

Step 3. (Apply Cauchy–Schwarz and L -smoothness.) For each $t \in [0, 1]$, by Cauchy–Schwarz,

$$[\nabla C(\theta + t(\phi - \theta)) - \nabla C(\theta)]^\top(\phi - \theta) \leq \|\nabla C(\theta + t(\phi - \theta)) - \nabla C(\theta)\| \cdot \|\phi - \theta\|.$$

By L -smoothness applied to the two arguments $\theta + t(\phi - \theta)$ and θ , which differ by $t(\phi - \theta)$,

$$\|\nabla C(\theta + t(\phi - \theta)) - \nabla C(\theta)\| \leq L t \|\phi - \theta\|.$$

Combining, the integrand is at most $L t \|\phi - \theta\|^2$, and so

$$C(\phi) - C(\theta) - \nabla C(\theta)^\top(\phi - \theta) \leq \int_0^1 L t \|\phi - \theta\|^2 dt = \frac{L}{2} \|\phi - \theta\|^2,$$

which is the Descent Lemma. ■

A simple consequence used in both convergence proofs below: applying the Descent Lemma with one gradient descent step $\phi = \theta - \alpha \nabla C(\theta)$ gives

$$C(\phi) \leq C(\theta) - \alpha \left(1 - \frac{\alpha L}{2}\right) \|\nabla C(\theta)\|^2,$$

and in particular, with $\alpha = 1/L$,

$$C(\theta - L^{-1} \nabla C(\theta)) \leq C(\theta) - \frac{1}{2L} \|\nabla C(\theta)\|^2. \quad (3.1)$$

Squared-gradient bound. If C is L -smooth with minimiser θ^* , then for all θ ,

$$\|\nabla C(\theta)\|^2 \leq 2L(C(\theta) - C(\theta^*)). \quad (3.2)$$

Proof: The Descent Lemma applied at the iterate $\theta - L^{-1} \nabla C(\theta)$ gives (3.1). Since $C(\theta^*) \leq C(\theta - L^{-1} \nabla C(\theta))$ (any value of C is at least the minimum), we get $C(\theta^*) \leq C(\theta) - \frac{1}{2L} \|\nabla C(\theta)\|^2$, which rearranges to (3.2). ■

2. Proof of Gradient Descent Convergence: Convex Case

Theorem (convex case). Let C be convex and L -smooth with minimiser θ^* . Gradient descent with step size $\alpha = 1/L$ satisfies

$$C(\theta^{(t)}) - C(\theta^*) \leq \frac{L \|\theta^{(0)} - \theta^*\|^2}{2t}.$$

Proof: Let $\theta^{(t+1)} = \theta^{(t)} - L^{-1}\nabla C(\theta^{(t)})$. Define the squared distance to the optimum and the function-value gap by

$$r_t := \|\theta^{(t)} - \theta^*\|^2, \quad \delta_t := C(\theta^{(t)}) - C(\theta^*) \geq 0.$$

Step 1. (One-step decrease.) From the one-step bound (3.1) (with $\theta = \theta^{(t)}$),

$$C(\theta^{(t+1)}) \leq C(\theta^{(t)}) - \frac{1}{2L} \|\nabla C(\theta^{(t)})\|^2.$$

In particular $\delta_{t+1} \leq \delta_t$, so δ_t is non-increasing.

Step 2. (Convexity lower bound.) Convexity of C gives the first-order inequality

$$C(\theta^*) \geq C(\theta^{(t)}) + \nabla C(\theta^{(t)})^\top (\theta^* - \theta^{(t)}),$$

which rearranges to

$$\nabla C(\theta^{(t)})^\top (\theta^{(t)} - \theta^*) \geq \delta_t.$$

Step 3. (Expand the distance to the optimum.) Squaring the update $\theta^{(t+1)} - \theta^* = (\theta^{(t)} - \theta^*) - L^{-1}\nabla C(\theta^{(t)})$,

$$r_{t+1} = r_t - \frac{2}{L} \nabla C(\theta^{(t)})^\top (\theta^{(t)} - \theta^*) + \frac{1}{L^2} \|\nabla C(\theta^{(t)})\|^2.$$

Step 4. (Combine.) Use Step 2 to bound the cross term and Step 1 to bound the squared-gradient term:

$$r_{t+1} \leq r_t - \frac{2\delta_t}{L} + \frac{2}{L} (C(\theta^{(t)}) - C(\theta^{(t+1)})) = r_t - \frac{2\delta_t}{L} + \frac{2(\delta_t - \delta_{t+1})}{L},$$

using $C(\theta^{(t)}) - C(\theta^{(t+1)}) = \delta_t - \delta_{t+1}$. The two δ_t terms cancel, leaving

$$r_{t+1} \leq r_t - \frac{2\delta_{t+1}}{L}, \quad \text{equivalently} \quad \delta_{t+1} \leq \frac{L}{2} (r_t - r_{t+1}).$$

Step 5. (Telescope.) Summing from $s = 0$ to $t - 1$ and using $r_t \geq 0$,

$$\sum_{s=1}^t \delta_s = \sum_{s=0}^{t-1} \delta_{s+1} \leq \frac{L}{2} (r_0 - r_t) \leq \frac{L}{2} \|\theta^{(0)} - \theta^*\|^2.$$

Step 6. (Monotonicity.) Step 1 gives $\delta_s \geq \delta_t$ for all $s \leq t$, so $t\delta_t \leq \sum_{s=1}^t \delta_s$. Combining with Step 5,

$$t\delta_t \leq \frac{L}{2} \|\theta^{(0)} - \theta^*\|^2,$$

which is the claimed rate. ■

3. Proof of Gradient Descent Convergence: Strongly Convex Case

Theorem (strongly convex case). *Let C be μ -strongly convex and L -smooth with unique minimiser θ^* . Gradient descent with step size $\alpha = 1/L$ satisfies*

$$\|\theta^{(t)} - \theta^*\|^2 \leq \left(1 - \frac{\mu}{L}\right)^t \|\theta^{(0)} - \theta^*\|^2.$$

Proof: Let $\theta^{(t+1)} = \theta^{(t)} - L^{-1}\nabla C(\theta^{(t)})$ and $r_t = \|\theta^{(t)} - \theta^*\|^2$.

Step 1. (Strong convexity lower bound on the cross term.) By definition of μ -strong convexity applied at $\theta = \theta^{(t)}$ and $\phi = \theta^*$,

$$C(\theta^*) \geq C(\theta^{(t)}) + \nabla C(\theta^{(t)})^\top (\theta^* - \theta^{(t)}) + \frac{\mu}{2} \|\theta^* - \theta^{(t)}\|^2.$$

Rearranging,

$$\nabla C(\theta^{(t)})^\top (\theta^{(t)} - \theta^*) \geq (C(\theta^{(t)}) - C(\theta^*)) + \frac{\mu}{2} r_t.$$

Step 2. (Squared-gradient bound.) By the lemma at the end of Section 1 (eq. 3.2),

$$\|\nabla C(\theta^{(t)})\|^2 \leq 2L(C(\theta^{(t)}) - C(\theta^*)).$$

Step 3. (Expand the distance to the optimum.) As in the convex proof,

$$r_{t+1} = r_t - \frac{2}{L} \nabla C(\theta^{(t)})^\top (\theta^{(t)} - \theta^*) + \frac{1}{L^2} \|\nabla C(\theta^{(t)})\|^2.$$

Step 4. (Combine.) Substituting the Step 1 bound on the cross term and the Step 2 bound on the squared-gradient term,

$$r_{t+1} \leq r_t - \frac{2}{L} \left[(C(\theta^{(t)}) - C(\theta^*)) + \frac{\mu}{2} r_t \right] + \frac{2}{L} (C(\theta^{(t)}) - C(\theta^*)),$$

and the two terms involving $C(\theta^{(t)}) - C(\theta^*)$ cancel, leaving

$$r_{t+1} \leq r_t - \frac{\mu}{L} r_t = \left(1 - \frac{\mu}{L}\right) r_t.$$

Step 5. (Iterate.) Apply Step 4 recursively from $t = 0$ to obtain $r_t \leq (1 - \mu/L)^t r_0$, which is the claimed rate. ■

4. Convexity of Logistic Regression

We prove that the binary cross-entropy loss for logistic regression is convex in the parameters, and identify a smoothness constant L .

Setup. Let $\sigma(u) = 1/(1 + e^{-u})$ denote the sigmoid (Week 1). With augmented inputs $\tilde{x}^{(i)} \in \mathbb{R}^{p+1}$ (a leading 1 for the bias, then the p features) and labels $y^{(i)} \in \{0, 1\}$, the binary cross-entropy training objective from Week 1 is

$$C(\theta) = -\frac{1}{n} \sum_{i=1}^n \left[y^{(i)} \log \sigma(\theta^\top \tilde{x}^{(i)}) + (1 - y^{(i)}) \log(1 - \sigma(\theta^\top \tilde{x}^{(i)})) \right].$$

Write $\hat{\sigma}_i := \sigma(\theta^\top \tilde{x}^{(i)})$ for the predicted probability that $y^{(i)} = 1$ at parameter θ , and let $\ell_i(\theta)$ denote the i -th summand above (so that $C(\theta) = \frac{1}{n} \sum_i \ell_i(\theta)$).

Per-sample Hessian. The chain rule gives the per-sample gradient

$$\nabla_{\theta} \ell_i = (\hat{\sigma}_i - y^{(i)}) \tilde{x}^{(i)},$$

using the standard identity $\sigma'(u) = \sigma(u)(1 - \sigma(u))$ together with $\partial \ell_{\text{BCE}} / \partial p = (p - y)/(p(1 - p))$, which cancel cleanly (this is the same calculation that produced the gradient formula for logistic regression in the *Gradients for Linear, Logistic, and SoftMax Regression* subsection of Week 1, §1.9 Gradient Descent). Differentiating $\hat{\sigma}_i - y^{(i)}$ once more with respect to θ ,

$$\nabla_{\theta}^2 \ell_i = \hat{\sigma}_i(1 - \hat{\sigma}_i) \tilde{x}^{(i)} \tilde{x}^{(i)\top}.$$

Convexity. Each $\nabla_{\theta}^2 \ell_i$ is a non-negative scalar times an outer product. For any $\phi \in \mathbb{R}^{p+1}$,

$$\phi^\top \nabla_{\theta}^2 \ell_i \phi = \hat{\sigma}_i(1 - \hat{\sigma}_i) (\phi^\top \tilde{x}^{(i)})^2 \geq 0,$$

so $\nabla_{\theta}^2 \ell_i \succeq 0$, i.e. each ℓ_i is convex. By the averaging property of finite-sum Hessians (Section 1.5 of the Week 3 notes), $\nabla^2 C(\theta) \succeq 0$ for all θ , so C is **convex**.

L -smoothness. Using $\hat{\sigma}_i(1 - \hat{\sigma}_i) \leq 1/4$ (the sigmoid output lies in $(0, 1)$, so $p(1 - p) \leq 1/4$),

$$\nabla^2 C(\theta) = \frac{1}{n} \sum_{i=1}^n \hat{\sigma}_i(1 - \hat{\sigma}_i) \tilde{x}^{(i)} \tilde{x}^{(i)\top} \preceq \frac{1}{4n} \sum_{i=1}^n \tilde{x}^{(i)} \tilde{x}^{(i)\top} = \frac{1}{4n} X^\top X,$$

where $X \in \mathbb{R}^{n \times (p+1)}$ is the design matrix from Week 1 (rows $\tilde{x}^{(i)\top}$). The largest eigenvalue of $\nabla^2 C(\theta)$ is therefore at most $\lambda_{\max}(X^\top X)/(4n)$, giving the smoothness constant

$$L = \frac{\lambda_{\max}(X^\top X)}{4n}.$$

Strong convexity. C is *not* strongly convex in general. The per-sample coefficient $\hat{\sigma}_i(1 - \hat{\sigma}_i)$ tends to zero as the prediction becomes very confident (i.e. $|\theta^\top \tilde{x}^{(i)}| \rightarrow \infty$), and the Hessian can shrink toward zero in those directions. So gradient descent on logistic regression achieves only the $O(1/t)$ rate of the convex case, unless an L2 penalty is added (which restores strong convexity with $\mu \geq \lambda$).

5. Convexity of Softmax Regression

We prove that the categorical cross-entropy loss for softmax regression is convex in the weight matrix.

Setup. With K classes, weight matrix $W \in \mathbb{R}^{K \times (p+1)}$, augmented inputs $\tilde{x}^{(i)} \in \mathbb{R}^{p+1}$, and one-hot labels $y^{(i)} \in \{0, 1\}^K$, define the softmax outputs at sample i by $\hat{y}^{(i)} = \text{softmax}(W\tilde{x}^{(i)})$, that is

$$\hat{y}_k^{(i)} = \frac{e^{(W\tilde{x}^{(i)})_k}}{\sum_{j=1}^K e^{(W\tilde{x}^{(i)})_j}}.$$

The categorical cross-entropy training objective from Week 1 is

$$C(W) = -\frac{1}{n} \sum_{i=1}^n \sum_{k=1}^K y_k^{(i)} \log \hat{y}_k^{(i)}.$$

Because $y^{(i)}$ is one-hot, only one term survives in the inner sum: writing $c^{(i)} \in \{1, \dots, K\}$ for the true class of sample i (so $y_{c^{(i)}}^{(i)} = 1$ and all other entries vanish), this simplifies to $C(W) = -\frac{1}{n} \sum_i \log \hat{y}_{c^{(i)}}^{(i)}$. Writing $z = W\tilde{x}^{(i)} \in \mathbb{R}^K$ for the per-sample pre-softmax scores and $h(z) := \log \sum_{k=1}^K e^{z_k}$ (the *log-sum-exp* function), the per-sample loss decomposes as

$$C_i(W) = -\log \hat{y}_{c^{(i)}}^{(i)} = -z_{c^{(i)}} + \log \sum_{k=1}^K e^{z_k} = h(z) - z_{c^{(i)}}.$$

Step 1: log-sum-exp is convex. A direct calculation of the gradient gives $\nabla h(z) = \text{softmax}(z)$, so the Hessian is

$$\nabla^2 h(z) = \text{Diag}(s) - s s^\top, \quad s := \text{softmax}(z).$$

To see this is positive semidefinite, observe that for any $\phi \in \mathbb{R}^K$,

$$\phi^\top (\text{Diag}(s) - s s^\top) \phi = \sum_{k=1}^K s_k \phi_k^2 - \left(\sum_{k=1}^K s_k \phi_k \right)^2.$$

This is exactly the variance of the random variable taking value ϕ_k with probability s_k (a categorical distribution on $\{1, \dots, K\}$ with probability vector s), which is always non-negative:

$$\phi^\top \nabla^2 h(z) \phi = \text{Var}_{k \sim s}[\phi_k] \geq 0.$$

So $\nabla^2 h(z) \succeq 0$ for all z , i.e. h is convex on $\mathbb{R}^K$.

Step 2: composition with a linear map preserves convexity. For any convex function $h : \mathbb{R}^K \rightarrow \mathbb{R}$ and any linear map $\mathcal{L} : \mathbb{R}^{K \times (p+1)} \rightarrow \mathbb{R}^K$ (linear in the entries of its matrix argument),

the function $g(W) := h(\mathcal{L}(W))$ is convex. Indeed, by linearity of $\mathcal{L}$, for any $\lambda \in [0, 1]$ and any W_1, W_2 ,

$$\begin{aligned} g(\lambda W_1 + (1 - \lambda)W_2) &= h(\mathcal{L}(\lambda W_1 + (1 - \lambda)W_2)) \\ &= h(\lambda \mathcal{L}(W_1) + (1 - \lambda)\mathcal{L}(W_2)) \\ &\leq \lambda h(\mathcal{L}(W_1)) + (1 - \lambda) h(\mathcal{L}(W_2)) \\ &= \lambda g(W_1) + (1 - \lambda) g(W_2), \end{aligned}$$

using convexity of h at the third line.

Step 3: assemble. The map $W \mapsto W\tilde{x}^{(i)}$ is linear in the entries of W , so by Step 2, $W \mapsto h(W\tilde{x}^{(i)})$ is convex in W . The map $W \mapsto -(W\tilde{x}^{(i)})_{c^{(i)}}$ is itself linear in W , hence affine and convex. Sums of convex functions are convex, so each

$$C_i(W) = h(W\tilde{x}^{(i)}) - (W\tilde{x}^{(i)})_{c^{(i)}}$$

is convex in W , and the average $C(W) = \frac{1}{n} \sum_i C_i(W)$ is convex by the same averaging property used in Section 4. So softmax regression is **convex**.

Strong convexity. As with logistic regression, $C(W)$ is not strongly convex without regularisation: at very confident predictions some s_k approach zero or one and $\nabla^2 h$ becomes rank-deficient. Adding L2 regularisation $\frac{\lambda}{2} \|W\|_F^2$ (sum of squared entries) restores λ -strong convexity.